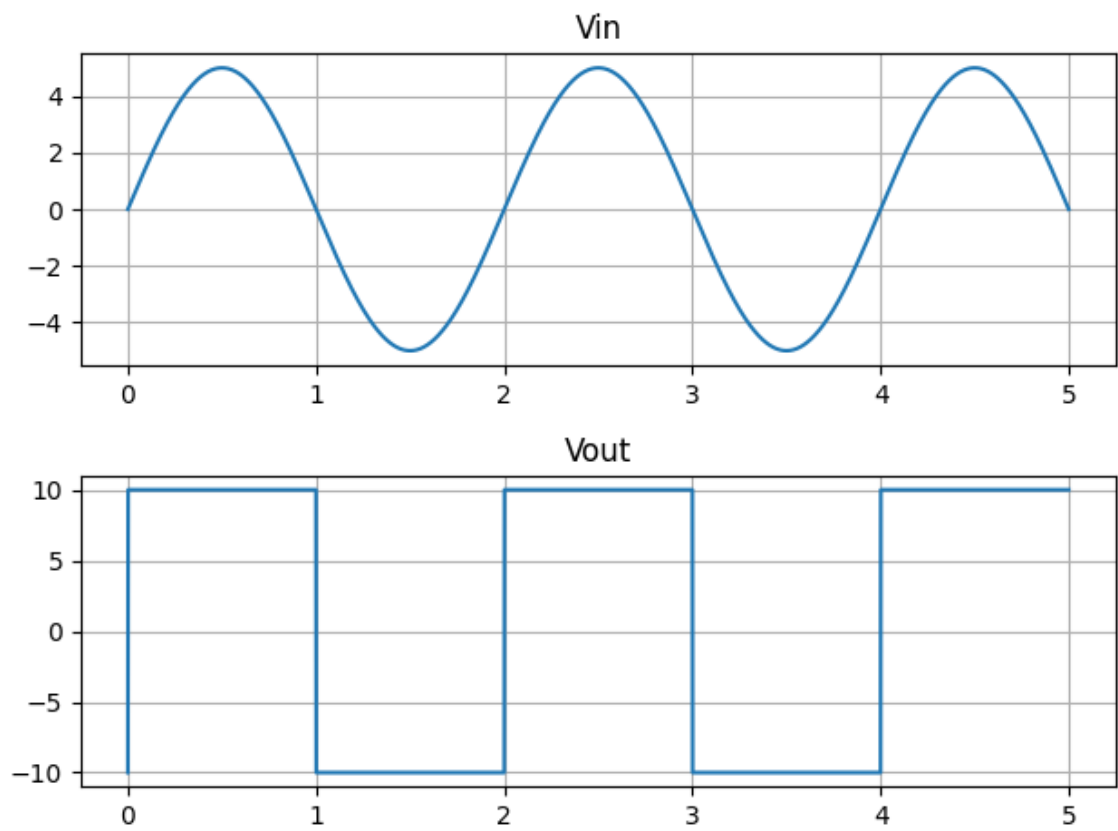


Exercise 2

Solution:

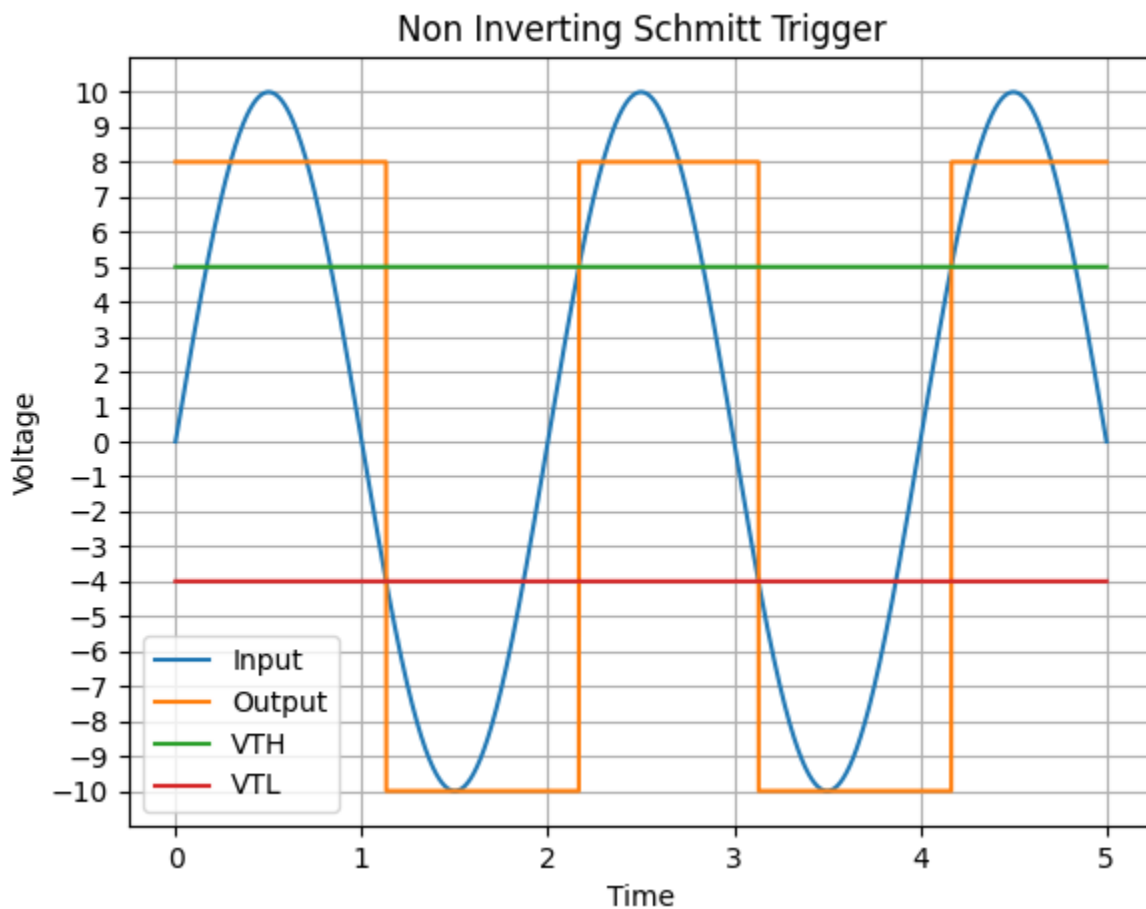


Exercise 3

Solution:

$$V_{TH} = -\frac{R_1}{R_3} \times V_L = -\frac{10}{20} \times -10 = 5 \text{ V}$$

$$V_{TL} = -\frac{R_1}{R_3} \times V_L = -\frac{10}{20} \times 8 = -4 \text{ V}$$



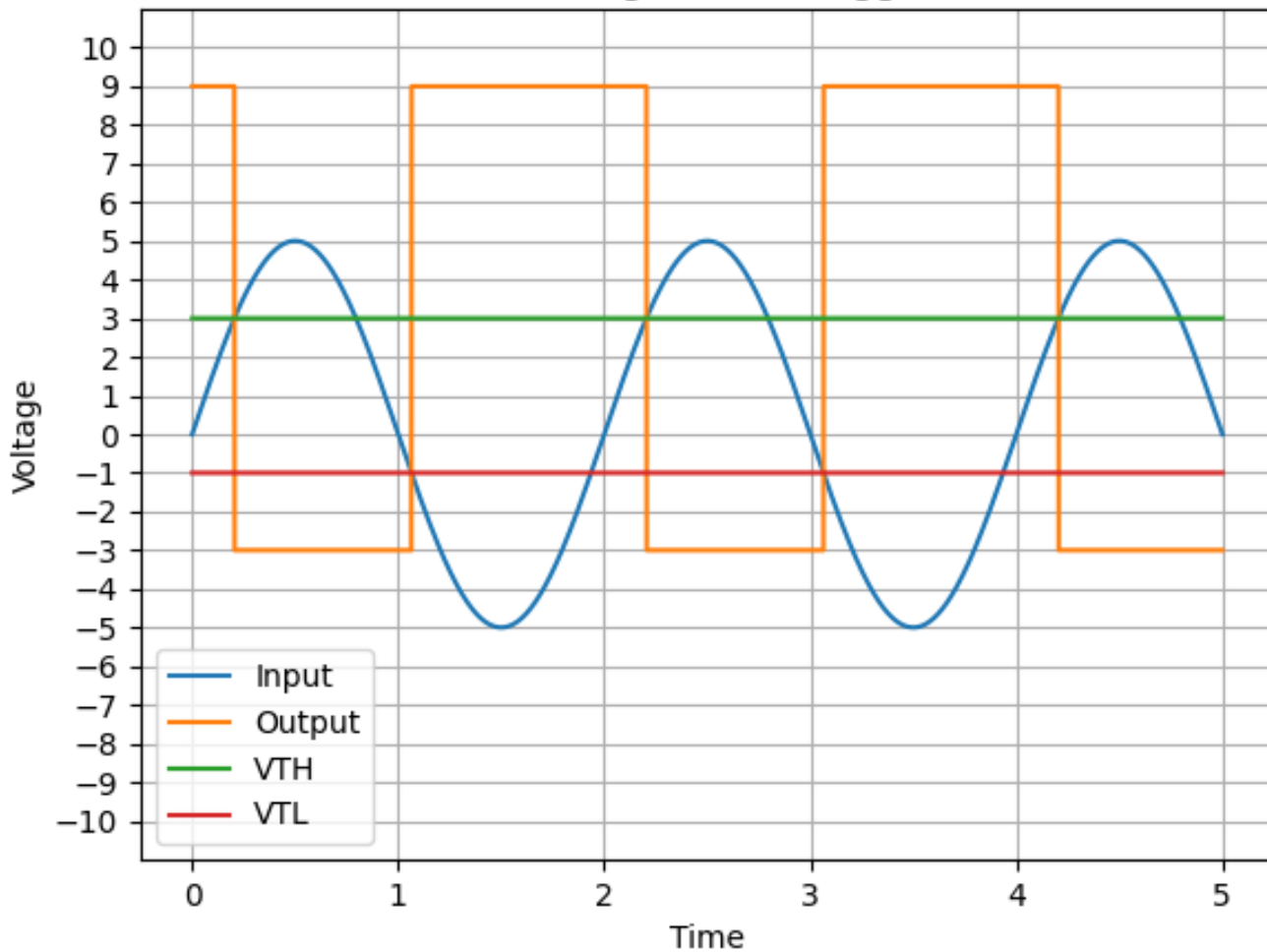
Exercise 4

Solution:

$$V_{TH} = \frac{R_1}{R_1 + R_3} \times V_H = \frac{10}{10 + 20} \times 9 = 3 \text{ V}$$

$$V_{TL} = \frac{R_1}{R_1 + R_3} \times V_L = \frac{10}{10 + 20} \times -3 = -1 \text{ V}$$

Inverting Schmitt Trigger



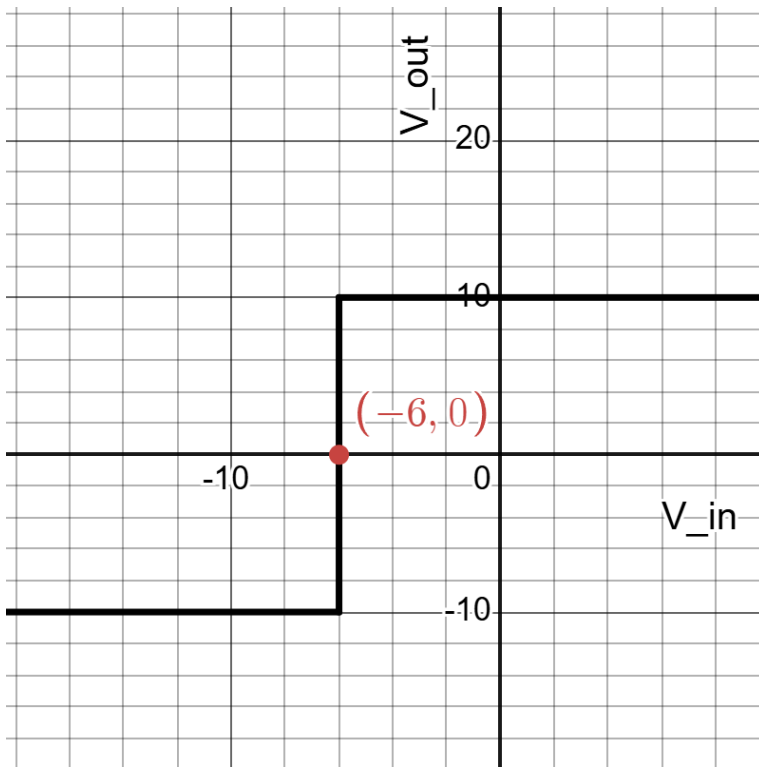
Exercise 5

Solution:

For input & reference voltage both applied to non-inverting terminal.

$$\text{If, } V_{in} > -\frac{R_2}{R_1} \times V_{ref} = -\frac{20}{10} \times 3 = -6 \text{ V, then } V_o = V_H$$

$$\text{If, } V_{in} < -\frac{R_2}{R_1} \times V_{ref} = -6 \text{ V, then } V_o = V_L$$



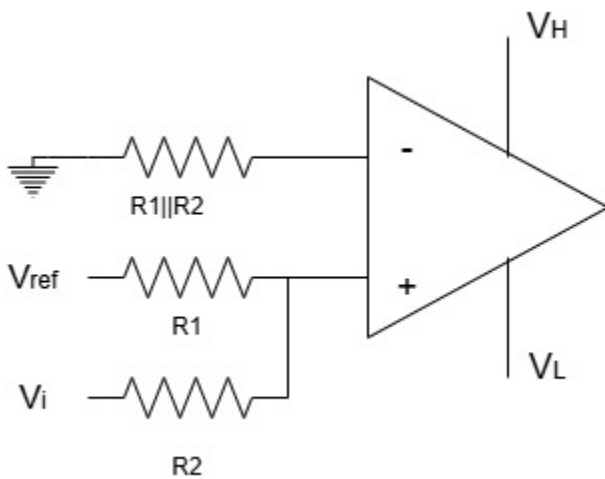
Exercise 6

Solution:

Assuming $R_1 = 5\text{ k}\Omega$ & $R_2 = 10\text{ k}\Omega$,

$$-\frac{R_2}{R_1} \times V_{ref} = 8 \rightarrow V_{ref} = -8 \times \frac{R_1}{R_2} = -8 \times \frac{5}{10} = -4\text{ V}$$

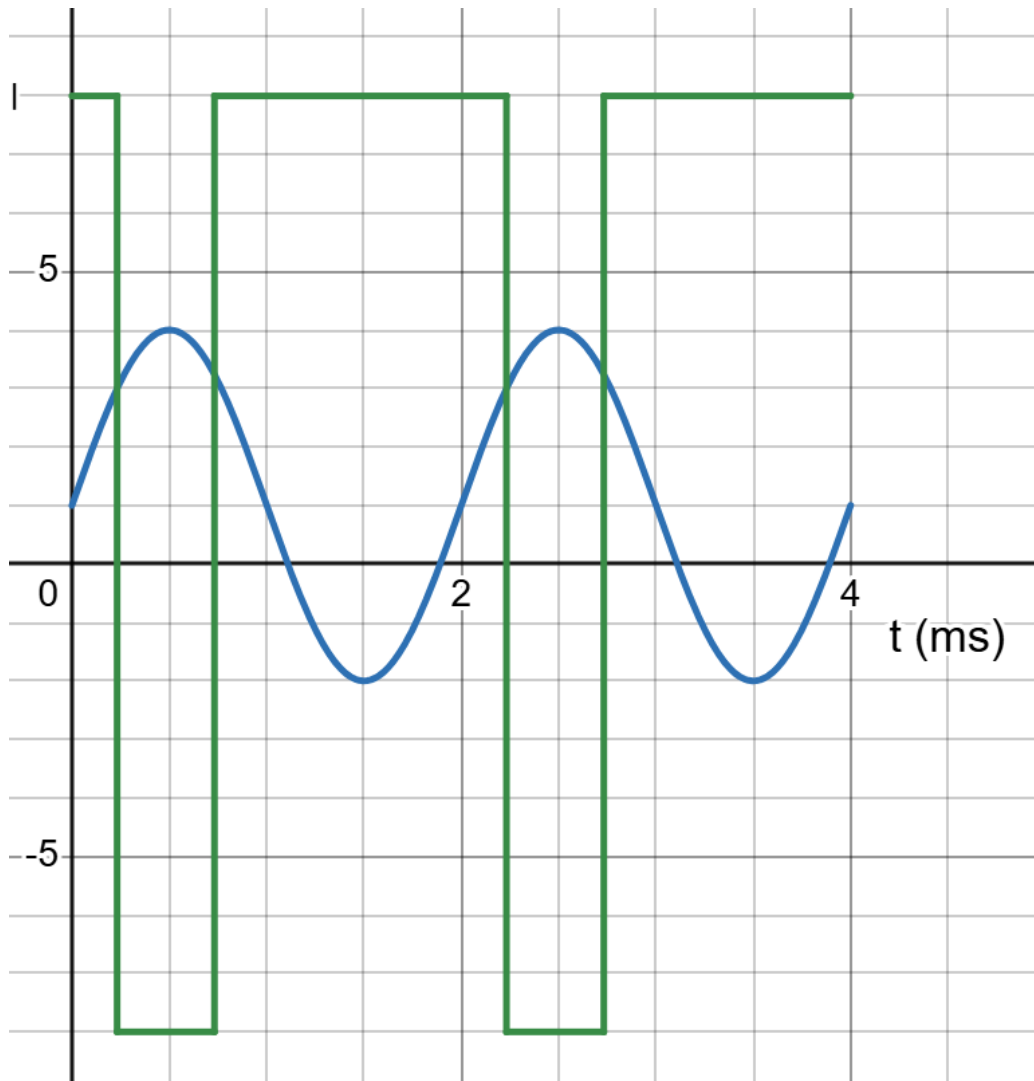
$$V_H = 11\text{ V}, V_L = -9.5\text{ V}$$



Exercise 7

Solution:

$$V_i < -\frac{R_2}{R_1} \times V_{ref} = -\frac{15}{10} \times -2 = 3 \text{ V}$$



The green graph is V_{out} vs time.

Exercise 8

Solution:

Non-Inverting:

$$V_{TH} = \frac{R_1+R_2}{R_2} \times V_{ref} - \frac{R_1}{R_2} \times V_L = \frac{5+20}{20} \times 3 - \frac{5}{20} \times -10 = 6.25 \text{ V}$$

$$V_{TL} = \frac{R_1+R_2}{R_2} \times V_{ref} - \frac{R_1}{R_2} \times V_H = \frac{5+20}{20} \times 3 - \frac{5}{20} \times 12 = 0.75 \text{ V}$$

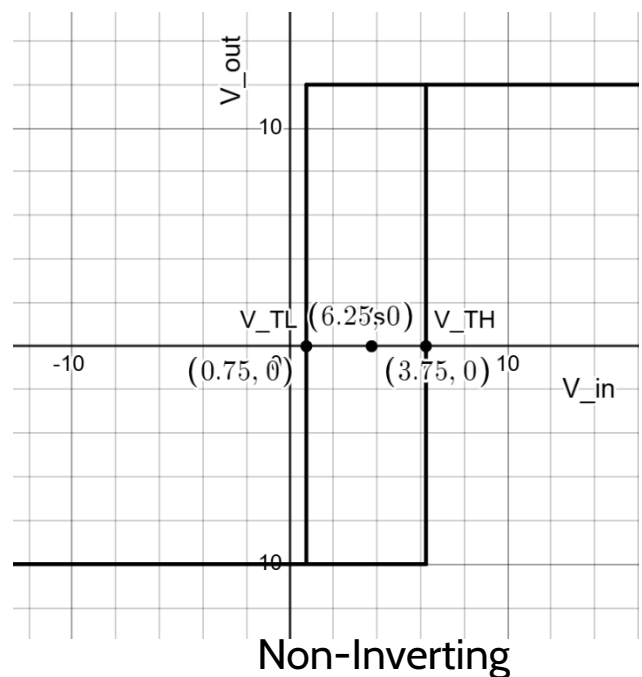
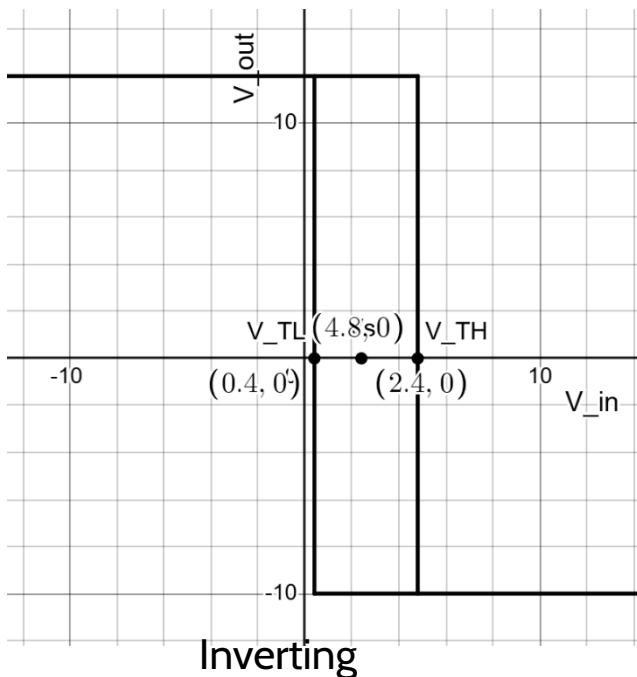
$$V_S = \frac{R_1+R_2}{R_2} V_{ref} = \frac{5+20}{5} \times 3 = 3.75 \text{ V}$$

Inverting:

$$V_{TH} = \frac{R_2}{R_1+R_2} \times V_{ref} + \frac{R_1}{R_1+R_2} \times V_H = \frac{5}{5+20} \times 3 + \frac{5}{5+20} \times 12 = 4.8 \text{ V}$$

$$V_{TL} = \frac{R_2}{R_1+R_2} \times V_{ref} + \frac{R_1}{R_1+R_2} \times V_L = \frac{5}{5+20} \times 3 + \frac{5}{5+20} \times -10 = 0.4 \text{ V}$$

$$V_S = \frac{R_2}{R_1+R_2} = 2.4 \text{ V}$$



Exercise 9

Solution:

From the VTC, $V_H = 10\text{ V}$, $V_L = -8\text{ V}$, $V_{TH} = 4\text{ V}$, $V_{TL} = -2.4\text{ V}$. Now,

$$V_{TH} = \frac{R_1}{R_1 + R_2} V_H + V_S \Rightarrow 4 = 10k + V_S \left[\text{Let } \frac{R_1}{R_1 + R_2} = k \right]$$

$$V_{TL} = kV_L + V_S \Rightarrow -2.4 = -8k + V_S$$

Solving the above two equations, $V_S = \frac{4}{9}\text{ V}$, $k = \frac{16}{45}$

$$k = \frac{16}{45} \Rightarrow \frac{R_1}{R_1 + R_2} = \frac{16}{45} \Rightarrow \frac{2}{2 + R_2} = \frac{16}{45} \\ \Rightarrow R_2 = 3.625\text{ k}\Omega$$

Now, $V_S = \frac{R_2}{R_1 + R_2} V_{ref}$. So,

$$\frac{4}{9} = \frac{3.625}{2 + 3.625} V_{ref} \Rightarrow V_{ref} = \frac{20}{29}\text{ V}$$

Thus, the designed Schmitt Trigger will be:

